Equiatomic AlCoCrFeNi High-Entropy Alloy

Complete Manual Paper Calculation • Phase Stability, Miedema Thermodynamics & Aqueous Corrosion Prediction

Target Alloy: $Al_{0.20}Co_{0.20}Cr_{0.20}Fe_{0.20}Ni_{0.20}$ (Equiatomic Al-Fe-Cr-Co-Ni, $x_i = 0.20$ each).
Key Characteristic: Benchmark prototype for Aluminum-strengthened HEAs. Unlike single-phase Cantor alloy, the large size mismatch of Al (143 pm) and strong negative Ni-Al / Co-Al enthalpies trigger **spinodal decomposition into BCC + ordered B2 (NiAl-type) phases**, altering both mechanical hardness and aqueous corrosion pathways.

1. Elemental Input Properties & Thermodynamic Database

Element (i)	at% (x_i)	M_i (g/mol)	r_i (pm)	VEC_i	Pauling χ_i	$T_{m,i}$ (K)	$E°$ (V)	n_i	Primary Oxide	$\Delta G°_{f,ox}$ (kJ/mol O_2)	PBR_i
Al	0.20	26.982	143	3	1.61	933	-1.662	3	Al_2O_3	-1055.0	1.28
Co	0.20	58.933	125	9	1.88	1768	-0.280	2	CoO	-428.4	1.76
Cr	0.20	51.996	128	6	1.66	2180	-0.744	3	Cr_2O_3	-705.4	2.07
Fe	0.20	55.845	126	8	1.83	1811	-0.447	2	Fe_2O_3	-494.8	2.14
Ni	0.20	58.693	124	10	1.91	1728	-0.257	2	NiO	-423.4	1.65

2. Stage I: Mass Balance & Stoichiometry

1. Mean Molar Mass (M):

$$\bar{M} = \Sigma\, x_i\, M_i = 0.20 \times (26.982 + 58.933 + 51.996 + 55.845 + 58.693) = 0.20 \times 252.449 = 50.490\ g/mol$$

2. Weight Fractions (w_i): $w_i = (x_i \times M_i / \bar{M}) \times 100\%$

- Al: (0.20 × 26.982 / 50.490) × 100% = **10.69 wt%** | Co: (0.20 × 58.933 / 50.490) × 100% = **23.34 wt%**
- Cr: (0.20 × 51.996 / 50.490) × 100% = **20.60 wt%** | Fe: (0.20 × 55.845 / 50.490) × 100% = **22.12 wt%**
- Ni: (0.20 × 58.693 / 50.490) × 100% = **23.25 wt%** (Total Sum = 100.00 wt%)

3. Stage II: Phase Stability & Miedema Thermodynamics

3. Valence Electron Concentration (VEC) & Guo Phase Rule:

$$VEC = \Sigma\, x_i\, VEC_i = 0.20 \times (3 + 9 + 6 + 8 + 10) = 0.20 \times 36 = 7.20$$

Guo Criterion: 6.87 ≤ VEC < 8.0 → **Duplex Microstructure (Disordered BCC + Ordered B2 / FCC)**.

DUPLEX PHASE REGIME

4. Ideal Configurational Mixing Entropy (ΔS_{mix}):

$$\Delta S_{mix} = -R\, \Sigma\, x_i \ln(x_i) = R \ln(5) = 8.314 \times 1.6094 = 13.38\ J/(mol \cdot K)$$

$\Delta S_{mix} \geq 1.5R$ (12.47 J/mol·K) → Full High-Entropy Alloy regime.

HIGH ENTROPY

5. Miedema Chemical Mixing Enthalpy (ΔH_{mix}):

Sum all 10 binary interaction pairs (ΔH_{ij} in kJ/mol):
- Al-Co: -19 | Al-Cr: -10 | Al-Fe: -11 | Al-Ni: -22 (Very strong B2 NiAl order)
- Co-Cr: -4 | Co-Fe: 0 | Co-Ni: 0 | Cr-Fe: -1 | Cr-Ni: -7 | Fe-Ni: -2

$\Sigma_{i<j} \Delta H_{ij}$ = (-19) + (-10) + (-11) + (-22) + (-4) + 0 + 0 + (-1) + (-7) + (-2) = **-76 kJ/mol**

$$\Delta H_{mix} = 4 \times \Sigma_{i<j}\, \Delta H_{ij}\, x_i\, x_j = 4 \times 0.04 \times (-76) = -12.16\ kJ/mol$$

High negative enthalpy indicates pronounced chemical ordering and strong intermetallic affinity.

6. Atomic Size Mismatch (δ) & Electronegativity Difference (Δχ):

$\bar{r}$ = 0.20 × (143 + 125 + 128 + 126 + 124) = 0.20 × 646 = **129.2 pm**
$\Sigma\, x_i (1 - r_i / \bar{r})^2$ = 0.20 × [(-0.1068)² + (0.0325)² + (0.0093)² + (0.0248)² + (0.0402)²] = 0.002958

$$\delta = \sqrt{[\ 0.002958\ ]} \times 100\% = 0.05439 \times 100\% = 5.44\%$$

$\bar{\chi}$ = 0.20 × (1.61 + 1.88 + 1.66 + 1.83 + 1.91) = **1.778** → **$\Delta\chi$ = 0.121**
Physical Meaning: The large atomic size of Al creates massive lattice strain (δ = 5.44% >> Cantor 1.12%), driving phase decomposition.

7. Average Melting Point (T_m) & Yang-Zhang Stability (Ω):

T_m = 0.20 × (933 + 1768 + 2180 + 1811 + 1728) = **1684.0 K**

$$\Omega = (T_m \times \Delta S_{mix}) / |\Delta H_{mix} \times 1000| = (1684.0 \times 13.38) / 12160 = 1.85$$

Zhang-Yang Criterion: Ω ≥ 1.1 and δ ≤ 6.6%. Although Ω = 1.85 > 1.1, the low value relative to its large negative ΔH_{mix} (-12.16 kJ/mol) proves that entropic stabilization cannot prevent the formation of ordered B2 (Ni,Al)-rich nanoprecipitates!

Ω = 1.85 (PRECIPITATION ACTIVE)

4. Stage III: Thermodynamic Passivation & Aqueous Corrosion Prediction

8. Modified High-Entropy Pitting Resistance Equivalent Number ($PREN_{HEA}$):

$$PREN_{ASTM} = wt\%\ Cr + 3.3\ (wt\%\ Mo + 0.5\ wt\%\ W) = 20.60 + 0 = 20.60$$

$$PREN_{HEA} = PREN_{ASTM} - 2.5 \times \max(0,\ wt\%\ Al - 3.0) = 20.60 - 2.5 \times (10.69 - 3.0) = 20.60 - 19.22 = 1.38$$

Mechanistic Rationale: In chloride media, Al contents > 3 wt% promote galvanic phase segregation. The Al-rich B2 phase undergoes preferential anodic dissolution (dealloying), drastically lowering pitting resistance.

$PREN_{HEA}$ = 1.38 (PITTING PRONE)

9. Ellingham Oxide Formation Affinity (ΔG_{ox}^{eff}):

$$\Delta G_{ox}^{eff} = \Sigma\ x_i\ \Delta G^{\circ}_{f,ox,i} = 0.20 \times [(-1055.0) + (-428.4) + (-705.4) + (-494.8) + (-423.4)] = -621.4\ kJ/mol\ O_2$$

Thermodynamic Order of Oxidation: Al_2O_3 (-1055.0 kJ/mol) > Cr_2O_3 (-705.4 kJ/mol) > Fe_2O_3 (-494.8 kJ/mol) > CoO (-428.4 kJ/mol) > NiO (-423.4 kJ/mol).
Al exhibits overwhelming oxygen affinity, driving outer Al_2O_3 film growth at high temperatures, but creating vulnerable passive boundaries in aqueous Cl^-.

10. Multicomponent Pilling-Bedworth Ratio (PBR_{eff}, Xu & Gao 2000):

$$PBR_{eff} = \Sigma\ x_i\ PBR_i = 0.20 \times (1.28 + 1.76 + 2.07 + 2.14 + 1.65) = 1.780$$

Criterion: 1.0 ≤ PBR_{eff} = 1.780 ≤ 2.10 → Oxide is under moderate protective compression without buckling or spallation.

PASSIVATING ENVELOPE

11. Micro-Galvanic Potential Spread (ΔE_{galv}) & Single-Phase Immunity Factor (I_{phase}):

Potential extreme: ΔE_{galv} = E°(Ni) - E°(Al) = -0.257 - (-1.662) = **1.405 V** (Extremely large electrochemical driving force)
Single-phase suppression factor:

$$I_{phase} = \min(1.0,\ \Omega\ /\ 1.1) \times (1.0 - \delta\ /\ 6.6\%) = 1.0 \times (1.0 - 5.44\ /\ 6.6) = 0.176\ \ (Only\ 17.6\%\ suppressed)$$

Microstructural Impact: In single-phase Cantor alloy, I_{phase} = 83%, extinguishing local cells. In AlCoCrFeNi, I_{phase} is only 17.6%! **82.4% of the micro-galvanic driving force remains active**, setting up intense galvanic cells between the BCC matrix (Cr, Fe rich) and the B2 precipitates (Ni, Al rich)!

SEVERE GALVANIC COUPLING

12. ASTM G102 Equivalent Weight (EW), Density (ρ), & Faraday Corrosion Rate (CR):

Dissolution valences: Al=3, Co=2, Cr=3, Fe=2, Ni=2.
Σ (w_i n_i / M_i) = (0.1069×3)/26.982 + (0.2334×2)/58.933 + (0.2060×3)/51.996 + (0.2212×2)/55.845 + (0.2325×2)/58.693 = **0.04753 eq/g**

$$EW = 1\ /\ 0.04753 = 21.04\ g/equivalent \quad | \quad Theoretical\ Density\ \rho \approx 6.73\ g/cm^3$$

In 3.5 wt% NaCl, experimental passive current density is i_{corr} ≈ 4.60 μA/cm² (due to micro-galvanic B2 dissolution):

$$CR_{mm/yr} = 3.27 \times 10^{-3} \times (i_{corr} \times EW)\ /\ \rho = 3.27e{-}3 \times (4.60 \times 21.04)\ /\ 6.73 = 0.0471\ mm/year$$

$$CR_{mpy} = 0.0471 \times 39.37 = 1.85\ mils/year \ \rightarrow\ PERFORMANCE:\ GRADE\ B\ (MODERATE\ /\ SELECTIVE\ LEACHING)$$

In 0.5 M H_2SO_4, i_{corr} increases to ~8.93 μA/cm² → **CR = 0.0910 mm/yr** (Kao et al. 2011, Materials Science & Engineering A).

5. Direct Benchmark: Cantor (CoCrFeMnNi) vs. AlCoCrFeNi

Descriptor	Cantor Alloy (CoCrFeMnNi)	Al-Containing HEA (AlCoCrFeNi)	Mechanistic Influence of Al Addition
Crystal Structure	Single FCC (A1)	**BCC + Ordered B2 Duplex**	Al destabilizes close-packed FCC → ordered B2
Valence Electrons (VEC)	8.00	**7.20**	VEC decreases across boundary 6.87 ≤ VEC < 8.0
Mixing Enthalpy (ΔH_{mix})	-3.20 kJ/mol	**-12.16 kJ/mol**	Strong Ni-Al (-22 kJ/mol) drives B2 intermetallic order
Size Mismatch (δ)	1.12%	**5.44%**	Al radius (143 pm) creates 4.8× higher lattice strain
Stability Parameter (Ω)	7.53	**1.85**	Entropy cannot overcome negative enthalpy ordering
$PREN_{HEA}$ Index	18.54	**1.38**	Severe penalty due to Al-induced phase separation
Galvanic Gap (ΔE_{galv})	0.928 V	**1.405 V**	Active Al (-1.662 V) creates massive potential gradient
Single-Phase Factor (I_{phase})	**0.830 (83% immune)**	**0.176 (82.4% vulnerable)**	Duplex boundaries sustain active micro-galvanic cells
Equivalent Weight (EW)	25.49 g/eq	**21.04 g/eq**	Higher dissolution efficiency per Faraday charge
Corrosion Rate (3.5% NaCl)	**0.0059 mm/yr (Grade A)**	**0.0471 mm/yr (Grade B)**	**~8× higher corrosion rate** due to dealloying of B2 phase

Key Takeaway for Alloy Design: While adding Al to CoCrFeNi dramatically increases yield strength and hardness (from ~150 MPa to >1100 MPa) via spinodal decomposition and B2 precipitate hardening, it creates a trade-off: **aqueous corrosion and pitting resistance decrease by nearly one order of magnitude** in chloride solutions.